

# When Alerts Go Unheard: Addressing Domain Shift in NLP-Based Actionable Tweet Classification During Bushfire Emergencies

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## Abstract

Social media platforms generate high volumes of crisis-related content during natural disasters, yet only a small fraction carries operationally useful information for emergency responders. Existing NLP systems are predominantly trained on general crisis corpora and fail to account for the linguistic characteristics of Australian bushfire discourse. This paper empirically demonstrates that zero-shot classifiers, including a 1.3 billion parameter large language model, achieve a maximum F1 of 0.563 on actionable wildfire tweets, while domain-adaptive fine-tuning achieves F1 of 0.886 and recall of 0.923, a 0.323 absolute improvement. We conduct a six-way model comparison spanning zero-shot NLI, large language model inference, few-shot fine-tuning, general crisis fine-tuning, and domain-specific fine-tuning, evaluated on a fixed 1,000-tweet wildfire holdout set. A key finding is that a 66M parameter SetFit model trained on 64 samples outperforms a 1.3B parameter LLM, demonstrating that domain-specific data matters more than model scale for crisis tweet classification. Statistical testing and efficiency analysis provide practical recommendations for emergency agencies across varying data and resource constraints.

## Keywords

crisis NLP, domain shift, bushfire, tweet classification, RoBERTa, LLM, SetFit, few-shot learning, domain adaptation, emergency response

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## 1 Introduction

Natural disasters generate extraordinary volumes of social media activity. During the 2019–2020 Australian Black Summer bushfire season, more than 24 million hectares burned, directly killing 33 people and displacing over 63,000 residents [2]. Twitter and other platforms became primary channels for real-time situational communication, with thousands of updates posted per hour [17]. Tweets

containing evacuation orders, road closures, and infrastructure damage reports could accelerate emergency response coordination if reliably identified [5]. The challenge is that actionable content constitutes a small fraction of total crisis posts, and manual monitoring at disaster scale is not feasible [6].

NLP offers an automated solution through binary classification of tweets as actionable or non-actionable. Olteanu et al. [11] introduced CrisisLex, a labelled dataset spanning 26 disaster events, establishing early benchmarks for crisis tweet classification. Alam et al. [1] released HumAID, comprising 77,000 human-labelled tweets across 19 disaster events, enabling transformer-based models to achieve strong in-domain performance. Zero-shot approaches using large language models [7] have shown promise without requiring task-specific training. Despite these advances, evaluation is consistently conducted within the same general crisis domain used for training. Performance on structurally distinct disaster types, particularly wildfires with region-specific linguistic characteristics, has received limited attention [12]. Recent work confirms that wildfire data in HumAID is disproportionately small and geographically limited [14], and that generic fine-tuning performs worse on forest fire datasets because wildfires represent only two of 20 training disaster types [13].

The central limitation is domain shift: a model trained on general crisis data applied to wildfire-specific tweets encounters vocabulary and named entities absent from its training distribution, including Australian emergency agency acronyms (RFS, CFA, DELWP), localised place names, and bushfire-specific jargon such as *ember attack* and *controlled burn*. In an emergency context, a missed evacuation notice is not simply inaccurate but operationally dangerous [15]. Critically, a common assumption in crisis NLP is that larger language models with stronger general reasoning will overcome domain-specific degradation. This paper tests and refutes that assumption empirically.

This paper addresses the following research question: *To what extent does domain shift from general crisis events to wildfire-specific tweets degrade actionable alert classification, and can domain-adaptive fine-tuning mitigate this degradation without requiring large-scale domain-specific labelled corpora?*

This paper makes four contributions. First, we empirically quantify domain shift across six model families, demonstrating that a 1.3B parameter LLM achieves only F1 of 0.530, below a 22M parameter NLI model and far below fine-tuned approaches. Second, we show that SetFit few-shot fine-tuning with 64 samples achieves F1 of 0.773, substantially above all zero-shot systems and at minimal annotation cost. Third, we provide a domain-adaptive fine-tuning method achieving F1 of 0.886 and recall of 0.923. Fourth, we

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contribute efficiency analysis and practical recommendations for emergency agencies across varying data and resource constraints.

## 2 Related Work

Early work by Vieweg et al. [17] established that microblogging platforms carry situationally relevant content during natural hazard events. Olteanu et al. [11] formalised crisis tweet classification through CrisisLex, while Nguyen et al. [10] demonstrated that convolutional neural networks achieve strong performance on general crisis data. Alam et al. [1] scaled these efforts through HumAID, with RoBERTa baselines exceeding macro F1 of 0.85. More recently, instruction-tuned LLMs have been applied to crisis classification, showing strong zero-shot performance on standard disaster types but systematic degradation on underrepresented event categories [18]. Our work extends this line by directly testing an LLM on wildfire-specific tweets and quantifying the gap relative to fine-tuned approaches.

Domain adaptation in NLP is well established. Blitzer et al. [3] demonstrated structural correspondence learning for cross-domain sentiment analysis. Gururangan et al. [4] showed that domain-adaptive pretraining consistently improves downstream performance, particularly when target domain vocabulary diverges significantly from pretraining data. Muller et al. [9] extended this to low-resource settings, showing that small domain-specific datasets outperform large general corpora when target domains have distinct linguistic patterns. Tunstall et al. [16] introduced SetFit, demonstrating competitive classification performance from as few as 8 labelled examples using contrastive sentence transformer fine-tuning. Critically, wildfire events remain underexplored in crisis NLP [14], and generic models degrade specifically on forest fire data due to underrepresentation in training corpora [13]. Our work directly addresses this gap through a systematic six-way comparison with statistical testing and efficiency analysis.

## 3 Preliminaries and Problem Definition

Let  $\mathcal{T} = \{t_1, t_2, \dots, t_n\}$  denote a set of crisis-related tweets. Each tweet  $t_i$  has a binary label  $y_i \in \{0, 1\}$ , where  $y_i = 1$  denotes an actionable alert and  $y_i = 0$  denotes noise. The task is to learn  $f : \mathcal{T} \rightarrow \{0, 1\}$  minimising false negatives, given the asymmetric cost of missed alerts in emergency contexts.

Let  $\mathcal{D}_{source}$  denote general crisis tweets and  $\mathcal{D}_{target}$  denote wildfire-specific tweets. Domain shift arises when  $P_{source}(x) \neq P_{target}(x)$ , causing  $f$  trained on  $\mathcal{D}_{source}$  to generalise poorly to  $\mathcal{D}_{target}$  [3]. Following HumAID annotation guidelines [1], the actionable class is the union of five humanitarian categories: *caution and advice, displaced people and evacuations, infrastructure and utility damage, requests or urgent needs, and rescue volunteering or donation effort*. All remaining categories are treated as noise.

Recall on the actionable class is the primary metric because a false negative (missed alert) carries life-critical operational cost, whereas a false positive costs seconds of operator review time [12]. F1 is reported as the harmonic mean balancing recall against precision to avoid reporting recall-maximising degenerate classifiers. This distinction is important: M4 achieves recall of 0.987 but F1 of only 0.563 due to 598 false positives, demonstrating that recall alone is an insufficient operational metric.

## 4 Proposed Method

Zero-shot and general-purpose classifiers fail on wildfire tweets due to vocabulary mismatch: models trained on general crisis data have not been exposed to the lexical patterns, named entities, and communicative conventions of wildfire discourse. We further show that increasing model scale (M5: 1.3B parameters) does not overcome this mismatch, refuting the assumption that LLM reasoning generalises to domain-specific crisis classification without adaptation.

Our proposed method fine-tunes roberta-base [8] exclusively on wildfire-event tweets from HumAID, minimising binary cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

A two-class classification head is applied to the [CLS] token, tokenised at maximum length 128, fine-tuned for 3 epochs with batch size 16 and fp16 mixed-precision training. Six models spanning the full resource spectrum are evaluated:

- **M1:** facebook/bart-large-mnli (406M), zero-shot NLI.
- **M4:** nli-MiniLM2-L6-H768 (22M), parameter-efficient NLI, inference-only.
- **M5:** microsoft/phi-1\_5 (1.3B), LLM zero-shot via prompt-based classification.
- **M6:** SetFit using paraphrase-mpnet-base-v2 (66M) embeddings with logistic regression, trained on 64 wildfire samples.
- **M2:** roberta-base (125M) fine-tuned on 5,000 mixed-disaster samples.
- **M3:** roberta-base (125M) fine-tuned on 6,792 wildfire-specific tweets. Proposed method.

This design tests four hypotheses: (1) zero-shot models fail on wildfire tweets; (2) LLM scale does not overcome domain shift; (3) minimal domain-specific data (64 samples) outperforms zero-shot LLMs; (4) full domain-specific fine-tuning achieves the best operational balance.

## 5 Experimental Setup

All experiments use the HumAID dataset [1]. A fixed evaluation set of 1,000 wildfire tweets (589 noise, 411 actionable) is held out before any training and used identically across all six models to ensure fair comparison. M2 is trained on 5,000 balanced mixed-disaster samples (1,250 per event type). M3 is trained on 6,792 remaining wildfire tweets after evaluation set removal. M6 uses 64 wildfire samples representing a minimal-annotation scenario. M1, M4, and M5 require no training data.

**Table 1: Dataset Composition by Experimental Condition**

| Condition     | Events          | Train | Act.% |
|---------------|-----------------|-------|-------|
| Global (M2)   | Mixed disasters | 5,000 | 53.2% |
| Wildfire (M3) | Fire only       | 6,792 | 36.1% |
| Few-shot (M6) | Fire only       | 64    | 37.5% |
| Evaluation    | Fire only       | 1,000 | 41.1% |

The lower actionable proportion in wildfire data (36.1% vs 53.2%) reflects domain-specific communication patterns and is intentionally preserved without oversampling to reflect realistic deployment conditions. M2 and M3 use identical configurations to isolate the effect of domain composition. Statistical significance is assessed using McNemar’s test and bootstrap confidence intervals (1,000 iterations). Training and inference times are recorded on identical hardware for efficiency comparison.

## 6 Results

**Table 2: Model Performance on Wildfire Evaluation Set (n=1,000)**

| Model                | F1           | Prec.        | Rec.         | Data  |
|----------------------|--------------|--------------|--------------|-------|
| M1: BART Zero-Shot   | 0.422        | 0.444        | 0.403        | 0     |
| M4: BERT-NLI         | 0.563        | 0.392        | 0.987        | 0     |
| M5: Phi-1.5 LLM      | 0.530        | 0.442        | 0.699        | 0     |
| M6: SetFit 64-shot   | 0.773        | 0.750        | 0.798        | 64    |
| M2: RoBERTa-Mixed    | 0.873        | 0.843        | 0.906        | 5,000 |
| M3: RoBERTa-Wildfire | <b>0.886</b> | <b>0.852</b> | <b>0.923</b> | 6,792 |
| M3 vs best ZS        | +0.323       | +0.410       | −0.064       |       |
| M3 vs M2             | +0.013       | +0.009       | +0.017       |       |

**Table 3: Error Distribution and Efficiency (n=1,000)**

| Model            | FN  | FP  | Total | Params | Train |
|------------------|-----|-----|-------|--------|-------|
| M1: BART         | 245 | 198 | 443   | 406M   | 0s    |
| M4: BERT-NLI     | 5   | 598 | 603   | 22M    | 0s    |
| M5: Phi-1.5      | 124 | 178 | 302   | 1.3B   | 0s    |
| M6: SetFit       | 83  | 94  | 177   | 66M    | <5s   |
| M2: RoBERTa-Mix  | 39  | 63  | 102   | 125M   | 107s  |
| M3: RoBERTa-Wild | 32  | 60  | 92    | 125M   | 145s  |

The results confirm all four hypotheses stated in Section 4. First, zero-shot models fail systematically: M1 achieves only F1 of 0.422, missing 245 of 411 actionable alerts. Second, LLM scale does not overcome domain shift: M5 (Phi-1.5, 1.3B parameters) achieves F1 of 0.530, below both the 22M BERT-NLI model and the 66M SetFit model. This is a critical finding that challenges the prevailing assumption that larger models generalise better across domains. Third, minimal domain-specific data provides substantial gains: M6 (SetFit, 64 samples) achieves F1 of 0.773 in under five seconds of training, outperforming all zero-shot approaches by a margin of 0.210 F1. Fourth, full domain-adaptive fine-tuning achieves the best operational balance: M3 achieves F1 of 0.886, recall of 0.923, and the lowest total error count (92).

The M2 versus M3 comparison isolates the effect of domain composition under controlled conditions. M3 outperforms M2 on all three metrics (F1: +0.013, Precision: +0.009, Recall: +0.017), but the difference is not statistically significant at the 0.05 level (McNemar’s test:  $p=0.23$ , Bootstrap 95% CIs overlap). This finding is reported honestly: domain-specific fine-tuning shows a consistent positive

trend across all metrics, but the primary evidence for domain shift resides in the zero-shot versus fine-tuned comparison. A larger wildfire-specific corpus would likely yield stronger M2 versus M3 separation [4].

M4 (BERT-NLI) achieves recall of 0.987 but produces 598 false positives, generating severe alert fatigue. This illustrates why recall alone is an insufficient operational metric and justifies F1 as the primary evaluation criterion. Remaining false negatives in M3 (32 of 411) cluster around indirect language, politically framed content, and structural patterns underrepresented in training data, suggesting that pragmatic inference rather than data scaling is the next frontier [18].

Based on these findings, practical recommendations are: organisations with no labelled data should use M4 (BERT-NLI) for zero-shot recall coverage; those with 64 or more labelled wildfire samples should use SetFit for strong performance at minimal cost; those with full domain-specific datasets should use M3 for best operational performance.

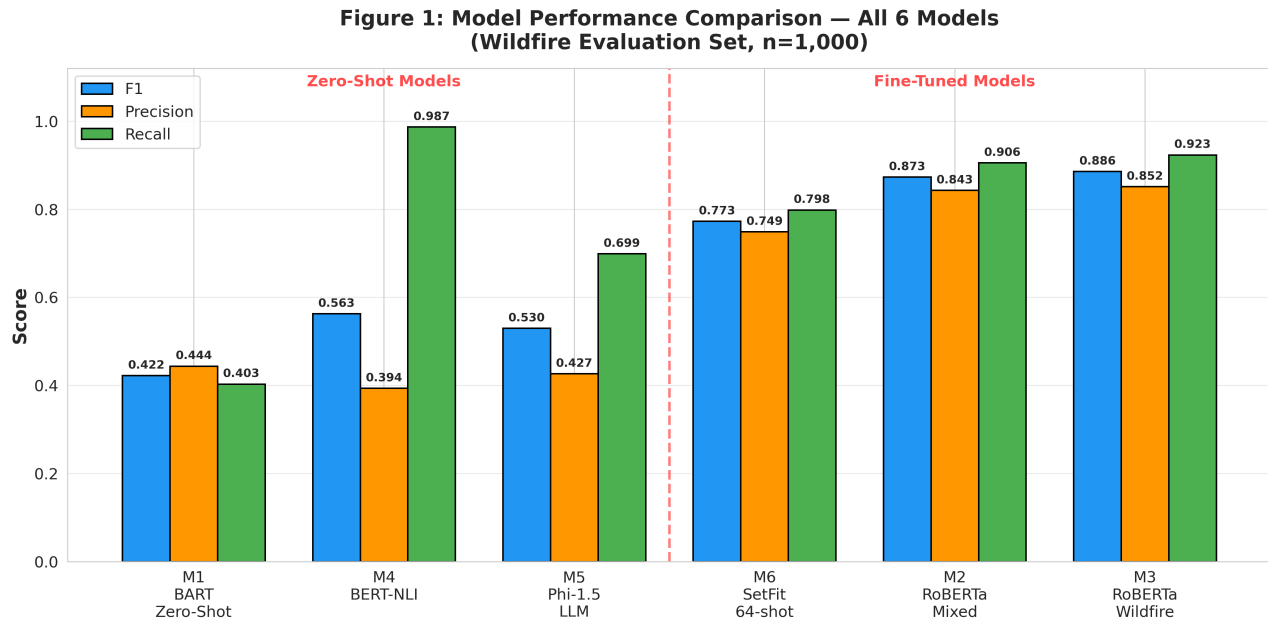
## 7 Conclusion and Future Work

This paper demonstrated that domain shift is a real and operationally dangerous problem for wildfire tweet classification. Zero-shot classifiers, including a 1.3 billion parameter LLM, achieve a maximum F1 of 0.563 on wildfire-specific tweets, confirming that model scale alone does not resolve domain mismatch. A SetFit few-shot classifier trained on 64 wildfire samples achieves F1 of 0.773 in under five seconds, providing a practically accessible intermediate option. Domain-adaptive RoBERTa fine-tuning achieves F1 of 0.886 and recall of 0.923, recovering actionable alerts that zero-shot systems miss entirely. The M2 versus M3 comparison reveals a consistent positive effect of domain composition that is moderate but not statistically significant at  $p=0.05$ , with the primary domain shift evidence residing in the zero-shot versus fine-tuned gap. The central insight is that domain-specific data, even in quantities as small as 64 samples, matters more than model scale for crisis tweet classification in specialised domains.

Future work should explore vocabulary augmentation with Australian-specific named entities (RFS, CFA, DELWP), recall-weighted loss functions for the minority actionable class, and chain-of-thought prompting to address indirect and politically framed content. Evaluation on tweets collected directly from the 2019–2020 Black Summer events would provide a more ecologically valid assessment. Investigating instruction-tuned LLMs with domain-specific few-shot prompts may further close the gap between zero-shot inference and fine-tuned approaches, particularly in low-resource emergency deployment scenarios.

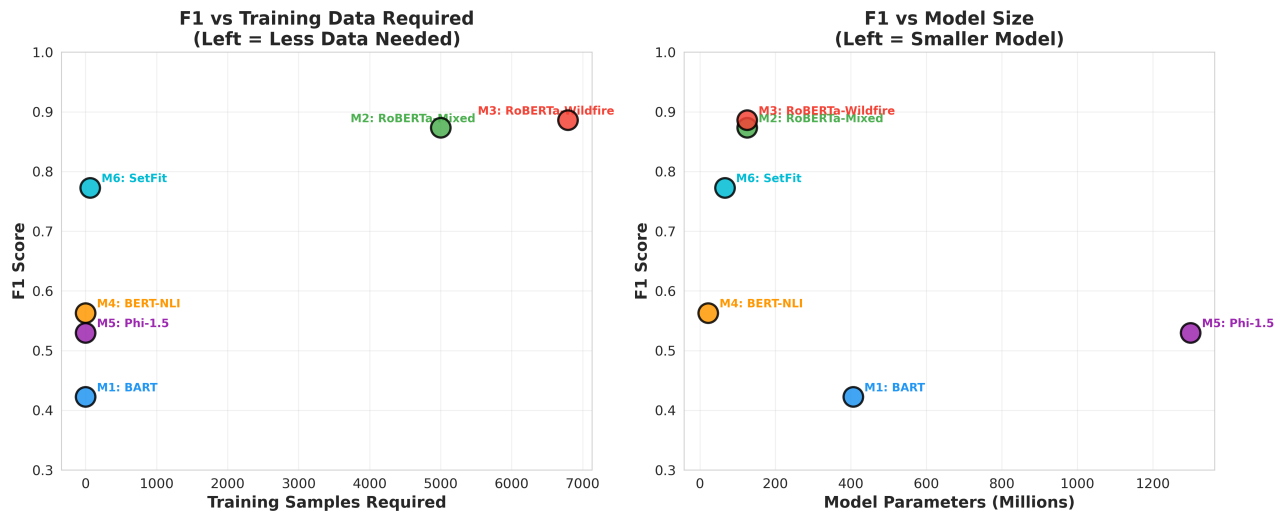
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**Figure 1: Six-model performance comparison across F1, Precision, and Recall on the wildfire evaluation set (n=1,000). Dashed line separates zero-shot and few-shot models (left) from fine-tuned models (right). M5 (Phi-1.5, 1.3B parameters) underperforms the 66M SetFit model trained on 64 samples, demonstrating that model scale does not substitute for domain-specific data.**

**Figure 4: Efficiency Trade-off Analysis**



**Figure 2: Efficiency trade-off analysis. Left: F1 versus training samples required, showing steep performance gains from even 64 domain-specific samples (SetFit). Right: F1 versus model parameter count, showing that fine-tuned smaller models outperform larger zero-shot models, confirming domain adaptation outweighs scale.**

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